

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

A rocket contained **467,000** kilograms (kg) of propellant before launch. Exactly **21** seconds after launch, **362,105** kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

Answer

- A. **4,995**
- B. **17,243**
- C. **39,481**
- D. **104,895**

Correct Answer: A

Rationale

Choice A is correct. It’s given that the rocket contained **467,000 kilograms (kg)** of propellant before launch and had **362,105 kg** remaining exactly **21** seconds after launch. Finding the difference between the amount, in **kg**, of propellant before launch and the remaining amount, in **kg**, of propellant after launch gives the amount, in **kg**, of propellant burned during the **21** seconds: **467,000 – 362,105 = 104,895**. Dividing the amount of propellant burned by the number of seconds yields $\frac{104,895}{21} = 4,995$. Thus, an average of **4,995 kg** of propellant burned each second after launch.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from finding the amount of propellant burned, rather than the amount of propellant burned each second.